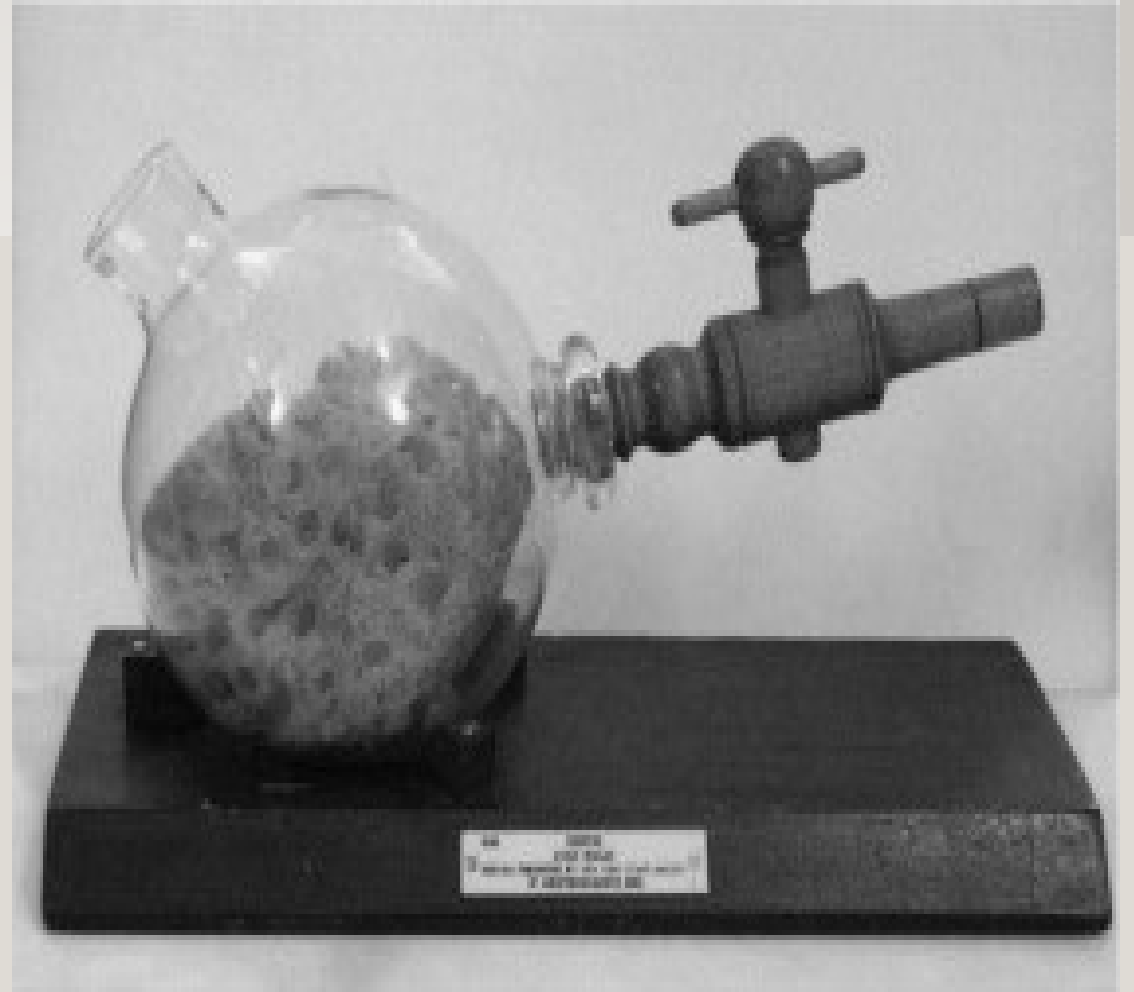


Anesthesia machine



The delivery system history 1846 – 76

- Morton's ether inhaler
- Is considered the first real anesthesia device.



Clovers chloroform apparatus

- In 1847, James Young Simpson introduced chloroform as an anesthetic.
- Used for the next 20 years
- In 1862 Thomas Clover invented a device that accurately measured and administered chloroform and air



CRITICAL REACTION OF THE LANCET

Ralph R. Frerichs

John Snow site

1818 London map

1846 London map

1856 Water map

1856 E&W map

1859 London map

1872 L region map

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After learning of the April 7th use of chloroform by Queen Victoria at the birth of Prince Leopold, *The Lancet* in an editorial challenged the accuracy of the information, and then condemned the Queen's physicians and Dr. John Snow (although not by name) for even considering anesthesia.

***The Lancet*, May 14, 1853.**

A very extraordinary report has obtained general circulation connected with the recent accouchement of her most gracious Majesty Queen Victoria. It has always been understood by the profession that the births of the Royal children in all instances have been unattended by any peculiar or untoward circumstances. Intense astonishment, therefore, has been excited throughout the profession by the rumor that her Majesty during her last labor was placed under the influence of chloroform, an agent which has unquestionably caused instantaneous death in a considerable number of cases. Doubts on this subject cannot exist. In several of the fatal examples persons in their usual health expired while the process of inhalation was proceeding, and the deplorable catastrophes were clearly and indisputably referable to the poisonous action of chloroform, and to that cause alone.



These facts being perfectly well known to the medical world, we could not imagine that any one had incurred the awful responsibility of advising the administration of chloroform to her Majesty during a perfectly natural labor with a seventh child (actually the eighth of her nine

Earlier anesthesia delivery systems

1. Open drop method



All Content

Images

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The American Journal of Nursing / Vol. 14, No. 4, Jan., 1914 / The Open Drop Meth

JOURNAL ARTICLE **OPEN ACCESS**

The Open Drop Method of Administering Ether

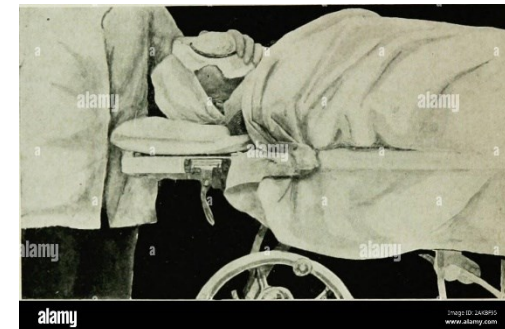
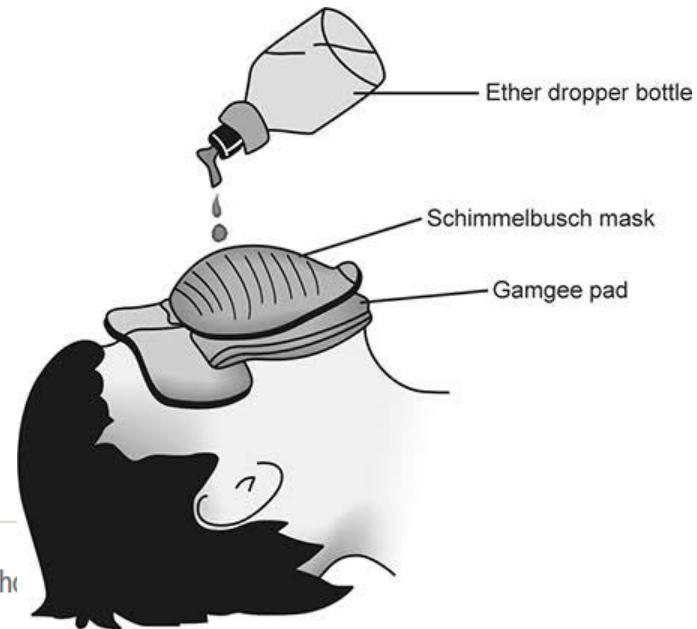
Mary A. Messer

The American Journal of Nursing, Vol. 14, No. 4 (Jan., 1914), pp.

260-263 (4 pages)

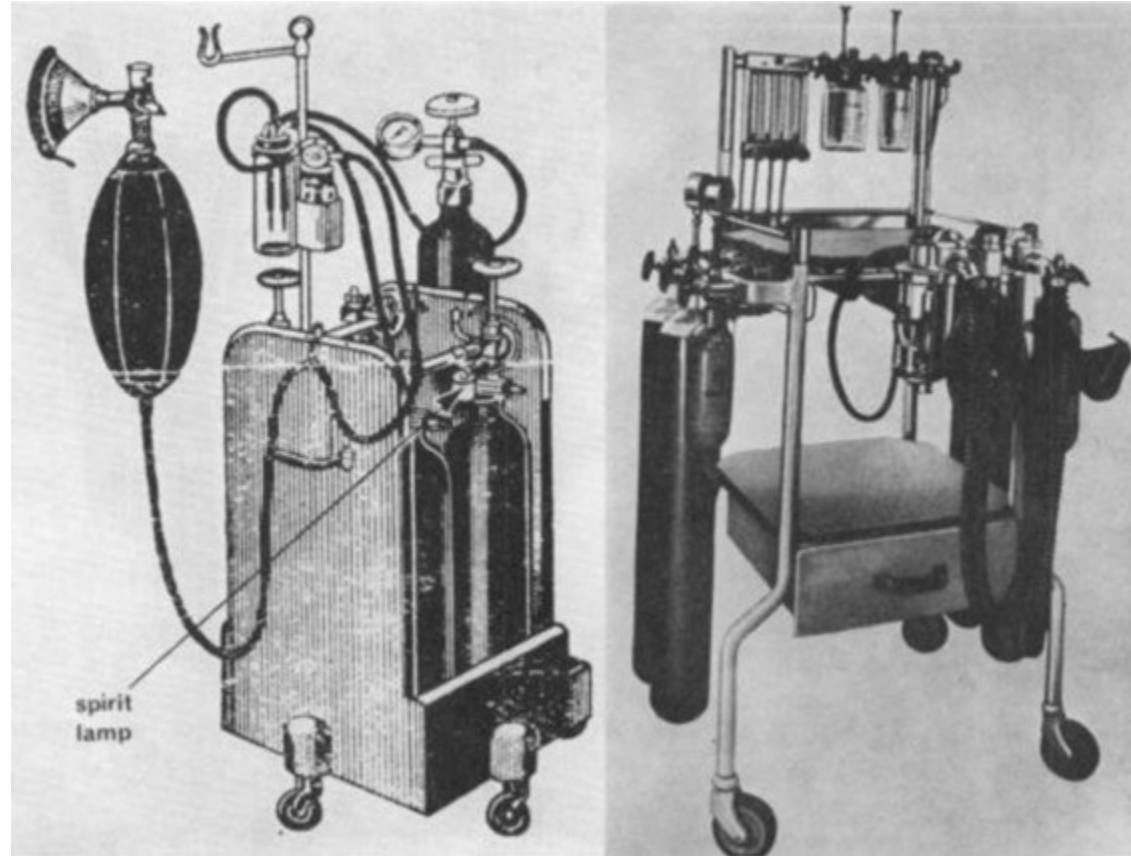
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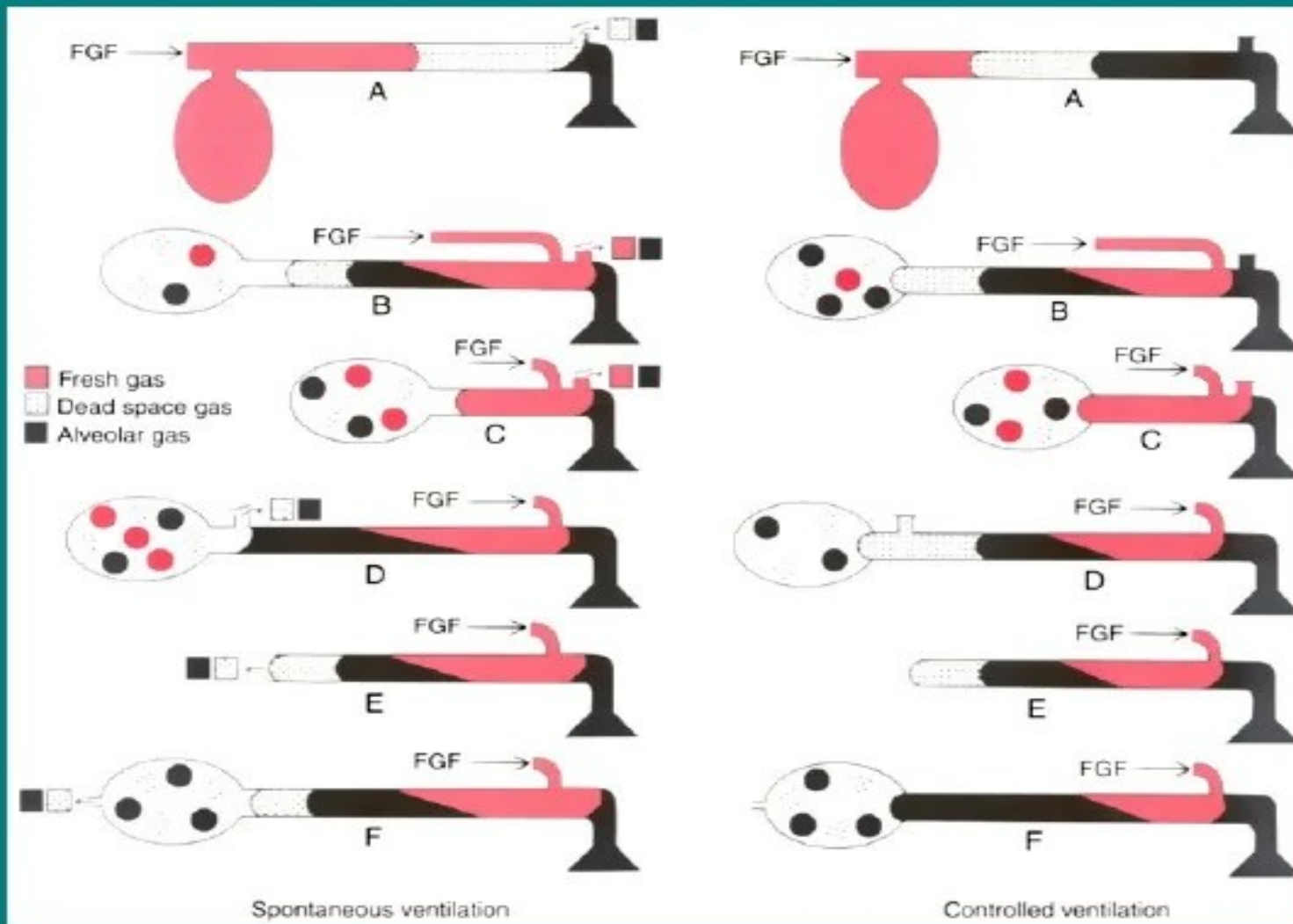
The precursor of standard anesthesia machine

- 1911 – bubble bottle as a measure of gas flow introduced – bubble rate used to estimate flow and proportion
- Shortly afterwards (1917) the **Boyle** machine was developed



Year	Development
1920	Addition of a chloroform vaporiser
1921-1924	Incorporation of the Waters carbon dioxide absorption system
1926	Introduction of bypass controls to regulate the quantity of ether/chloroform
1927	Addition of a carbon dioxide flow meter
1930	Inclusion of a plunger device in the vaporiser
1931-1933	Substitution of water flow meters by dry flow meters
1937	Replacement of dry flow meters with flowmeters
1941	Incorporation of the “Coxeter-Mushin Mark I” circle absorber unit in Boyle’s EMS model.
1952	Incorporation of non-interchangeable plugs in the gas connections.
1958	Incorporation of the Bodok seal in gas connections.

Mapleson Systems

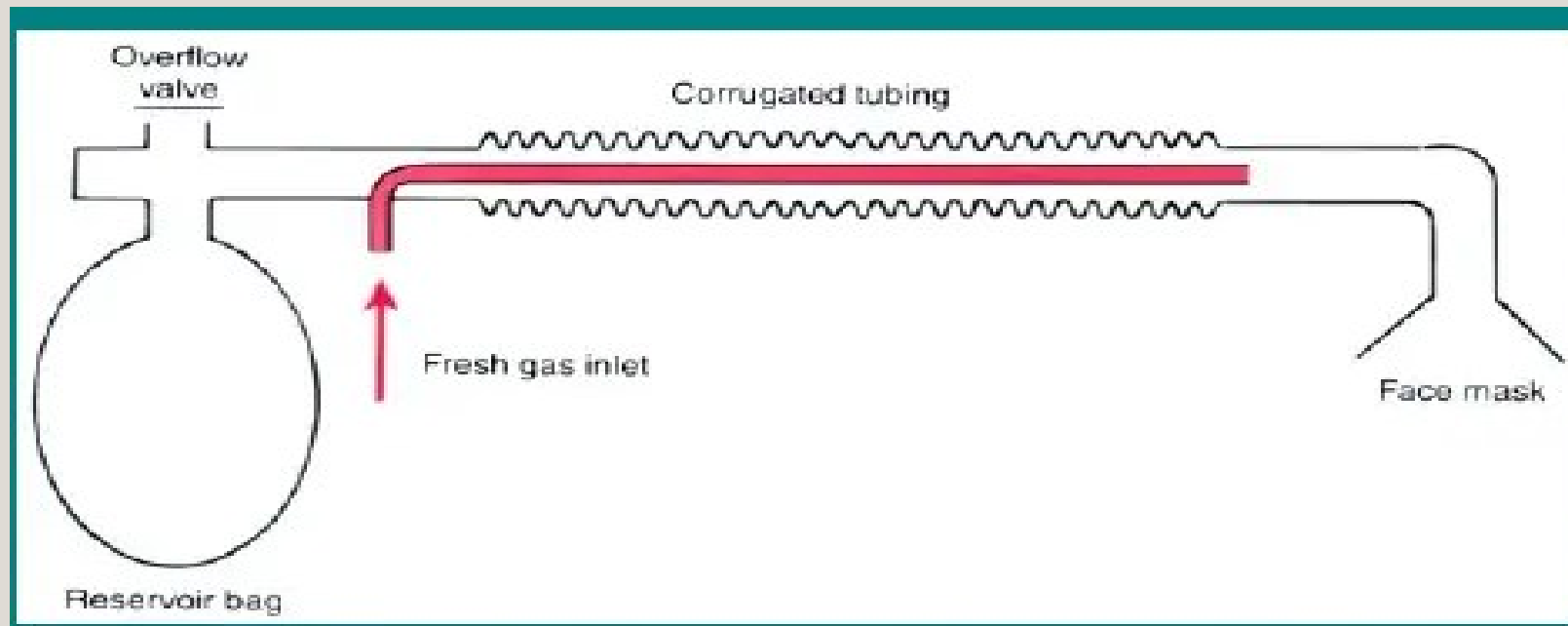


MAPLESON SYSTEM

- Prevention of rebreathing during spontaneous ventilation; $A > DFE > CB$
- During controlled ventilation $DFE > BC > A$
- A , B and C are rarely used today

THE BAIN CIRCUIT

- A modification Mapleson D system
- Spontaneous and controlled ventilation



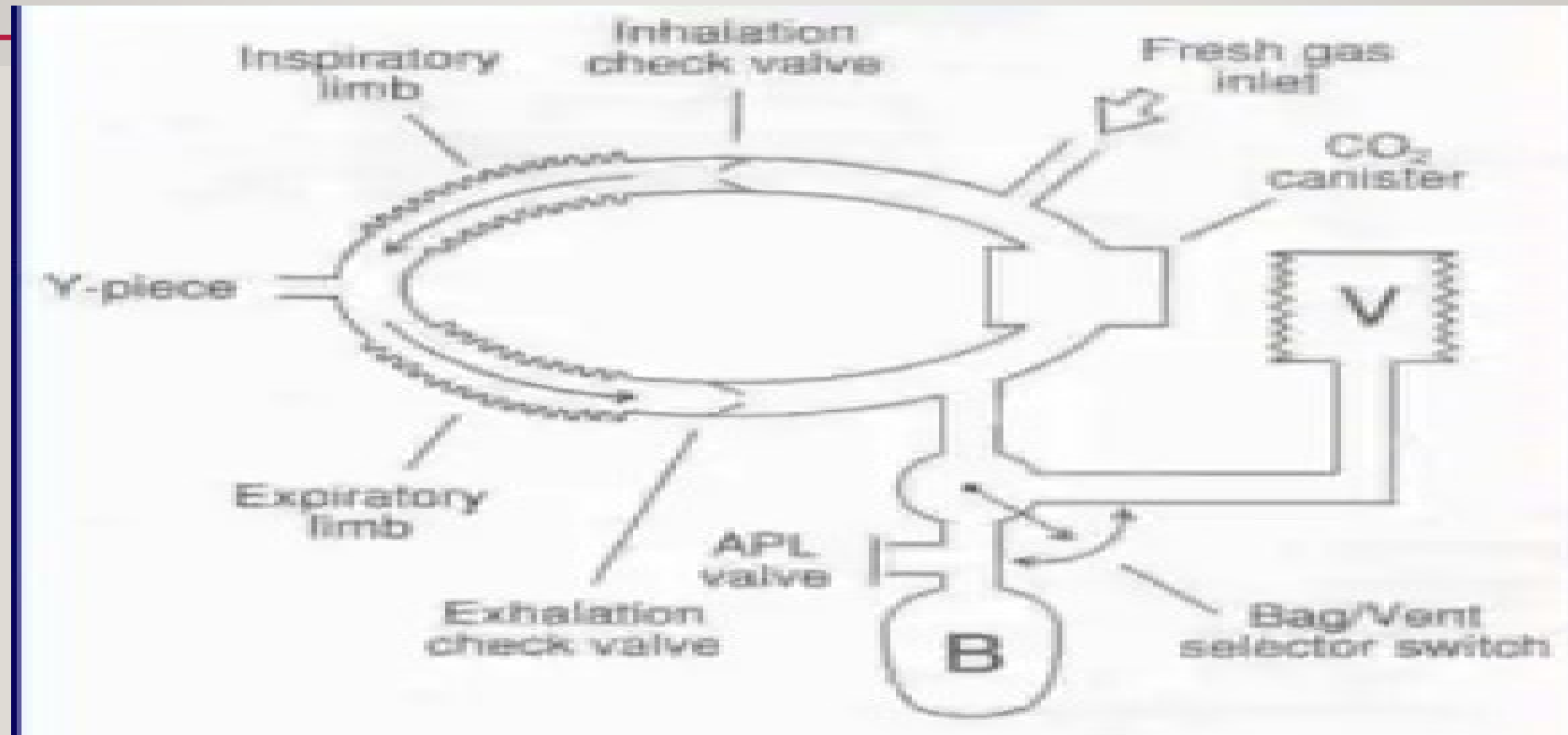
CIRCLE BREATHING SYSTEM

- A circle system can be semi open ,semi closed or closed depending on the amount of fresh gas inflow
- Semi open system has no rebreathing and requires a very high flow of fresh gas
- Semi closed system is associated with rebreathing of gases
- Closed system is one in which the inflow gas exactly matches that being consumed by the patient .

COMPONENT OF CIRCLE BREATHING SYSTEM

- A fresh gas inflow source
- Inspiratory and expiratory unidirectional valve
- Inspiratory and expiratory corrugated tubes and a Y piece connector
- An overflow or pop of valve referred to us APL valve
- A reservoir bag
- A canister containing carbon dioxide absorbent

CIRCLE SYSTEM



.....ANESTHESIA MACHINE

- ✓ Anaesthesia machine introduced by Gwathmey and Boyle nearly 100 years back has had many changes to its original design, mainly to improve the patient safety. However, still the basic structure remains the same.
- ✓ The basic function of an anesthesia machine is to prepare a gas mixture of precisely known, but variable composition.
- ✓ In its most basic form, the anesthesia machine receives medical gases from a gas supply, controls the flow and reduces the pressure of desired gases to a safe level, vaporizes volatile anesthetics into the final gas mixture, and delivers the gases to a breathing circuit connected to the patient's airway.
- ✓ Anesthesia machine; regardless of their manufacture, consist of the same basic components.

FUNCTIONS OF ANAESTHESIA MACHINE

The machine performs four essential functions:

1. Provides O₂
2. Accurately mixes anesthetic gases and vapors,
3. Enables patient ventilation
4. Minimizes anesthesia related risks to patients and staff.

Anesthesia delivery system

- Traditionally the pneumatic part of anesthesia machines are divided in to
 - High pressure
 - Intermediate
 - Low pressure systems

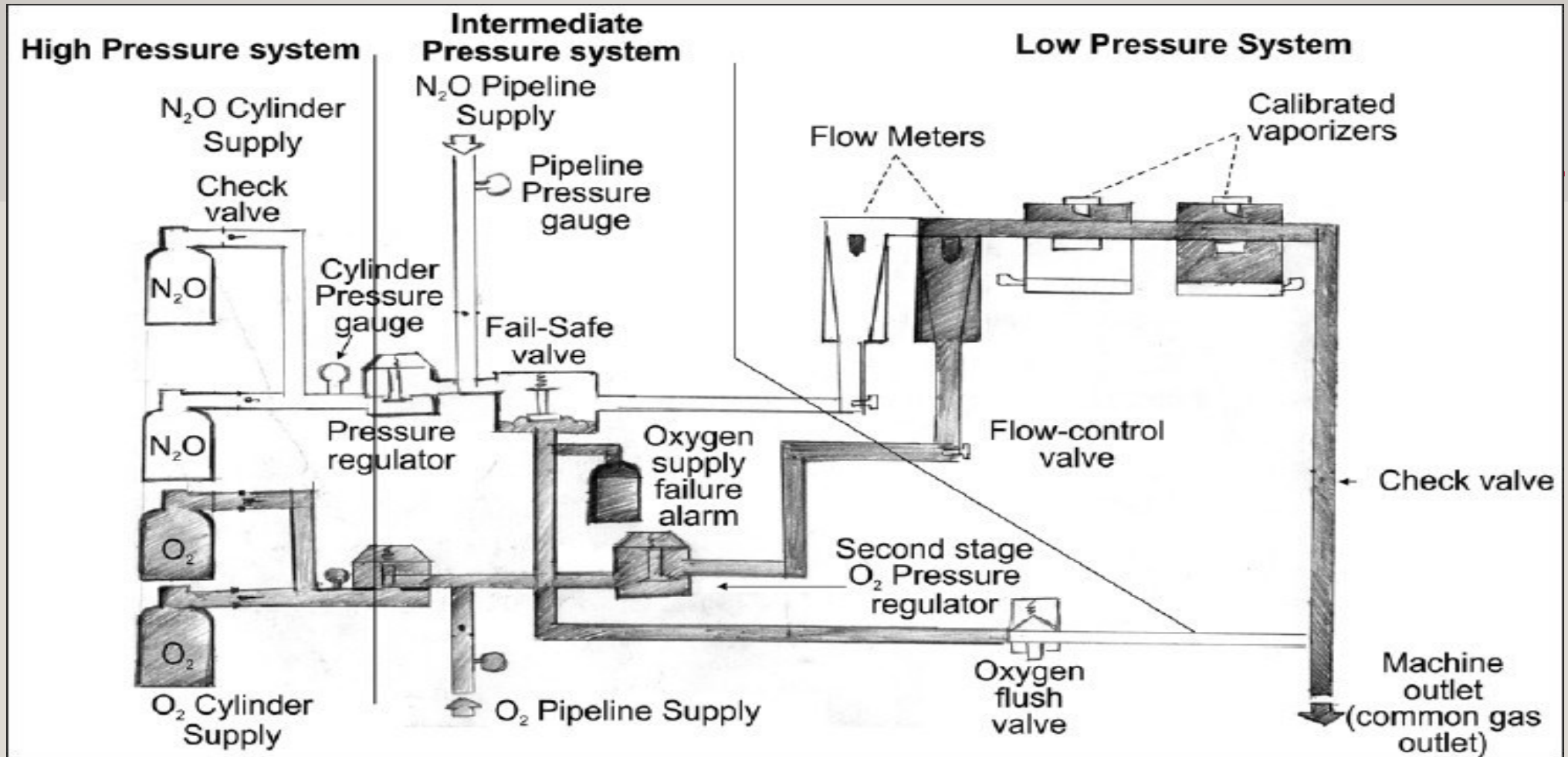
THE HIGH PRESSURE SYSTEM

The high pressure system consists of all parts of the machine, which receive gas at cylinder pressure. These include the following:

- (a) The hanger yoke which connects a cylinder to the machine(reserve gas cylinder holder)
- (b) The yoke block, used to connect cylinders larger than size E or pipeline hoses to the machine through the yoke,
- (c) The cylinder pressure gauge, which indicates the gas pressure in the cylinder
- (d) The pressure regulator, which converts a high variable gas pressure into a lower, more constant pressure, suitable for use in the machine.

Receives gases from the high pressure E cylinder attached to the back of anesthesia machine (2200)psig for oxygen ,745 psig for N₂O)

Check valve(prevent reverse flow of gas)

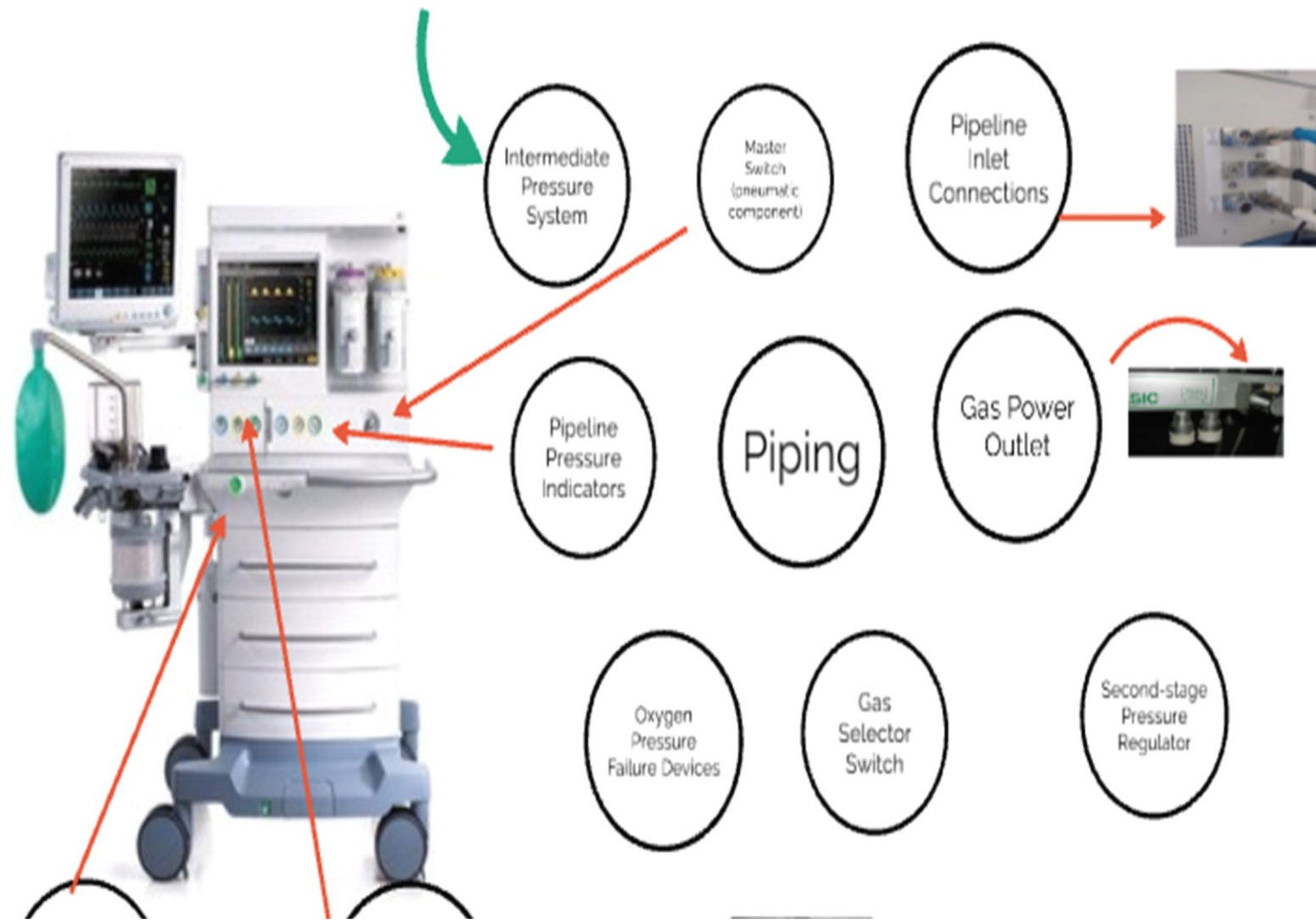


INTERMEDIATE PRESSURE SYSTEM

- Receives gases from the regulator or the hospital pipeline at pressure of 40 to 55psig

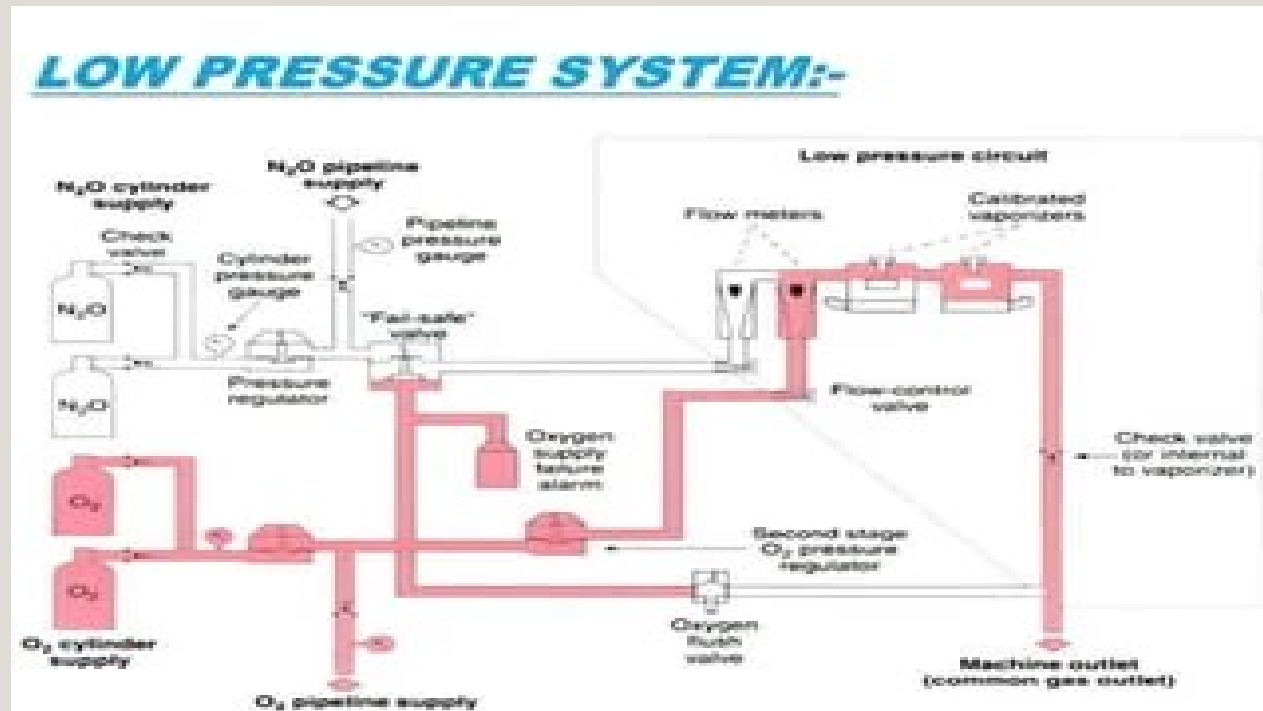
Consist of

- Pipeline inlet connection
- Pipeline pressure indicators
- Piping Gas power outlet
- Master switch
- Oxygen pressure failure device
- Oxygen flush
- Additional reducing device
- Flow control valves

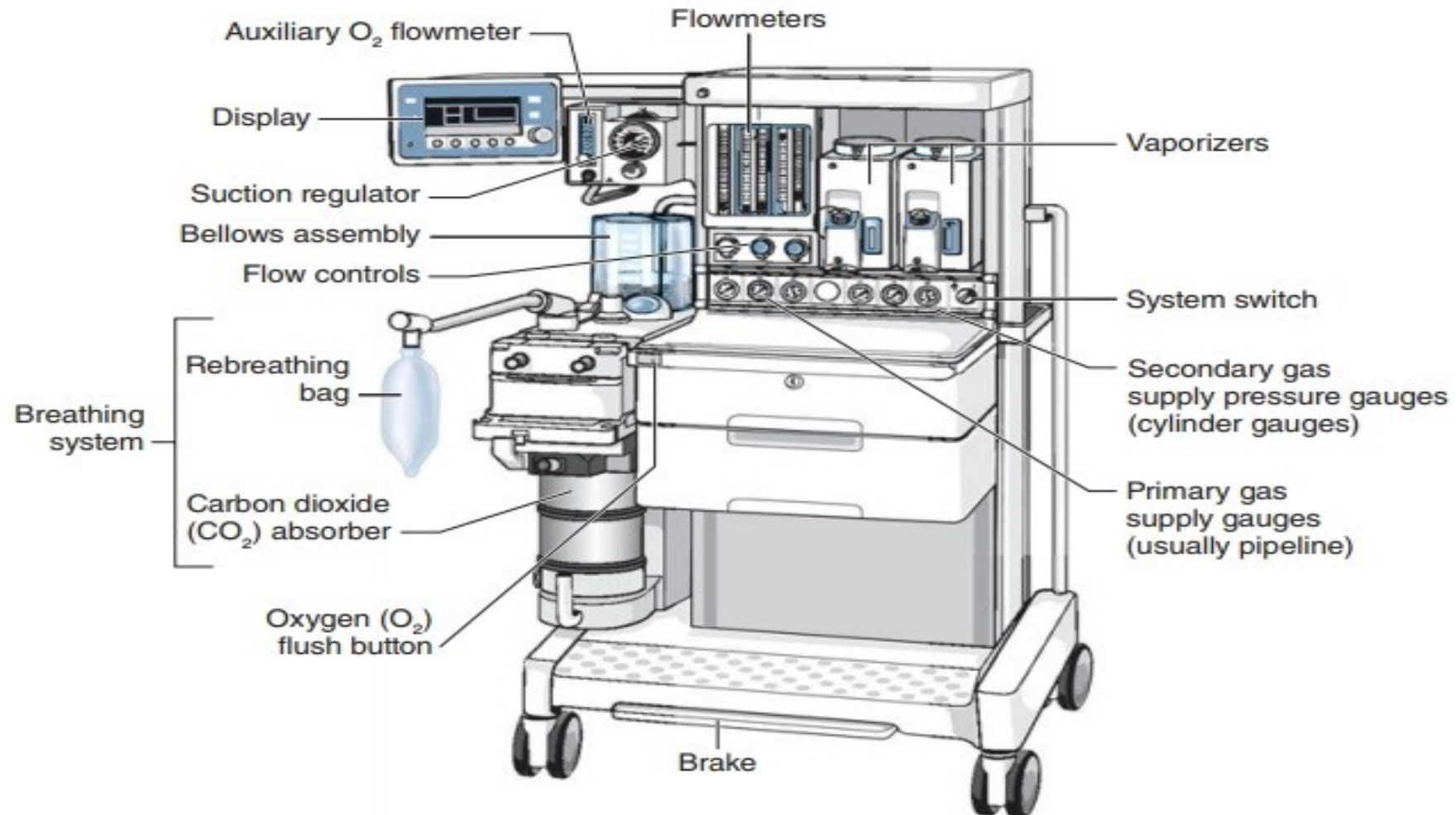


LOW PRESSURE SYSTEM

- The low pressure system is the part of the machine downstream of the flow meters in which the pressure is slightly above the atmosphere. The components in this system are:
 - Vaporizers mounted on the back bar
 - Back pressure safety devices
 - The common gas outlet.



MODERN ANESTHESIA MACHINE



B

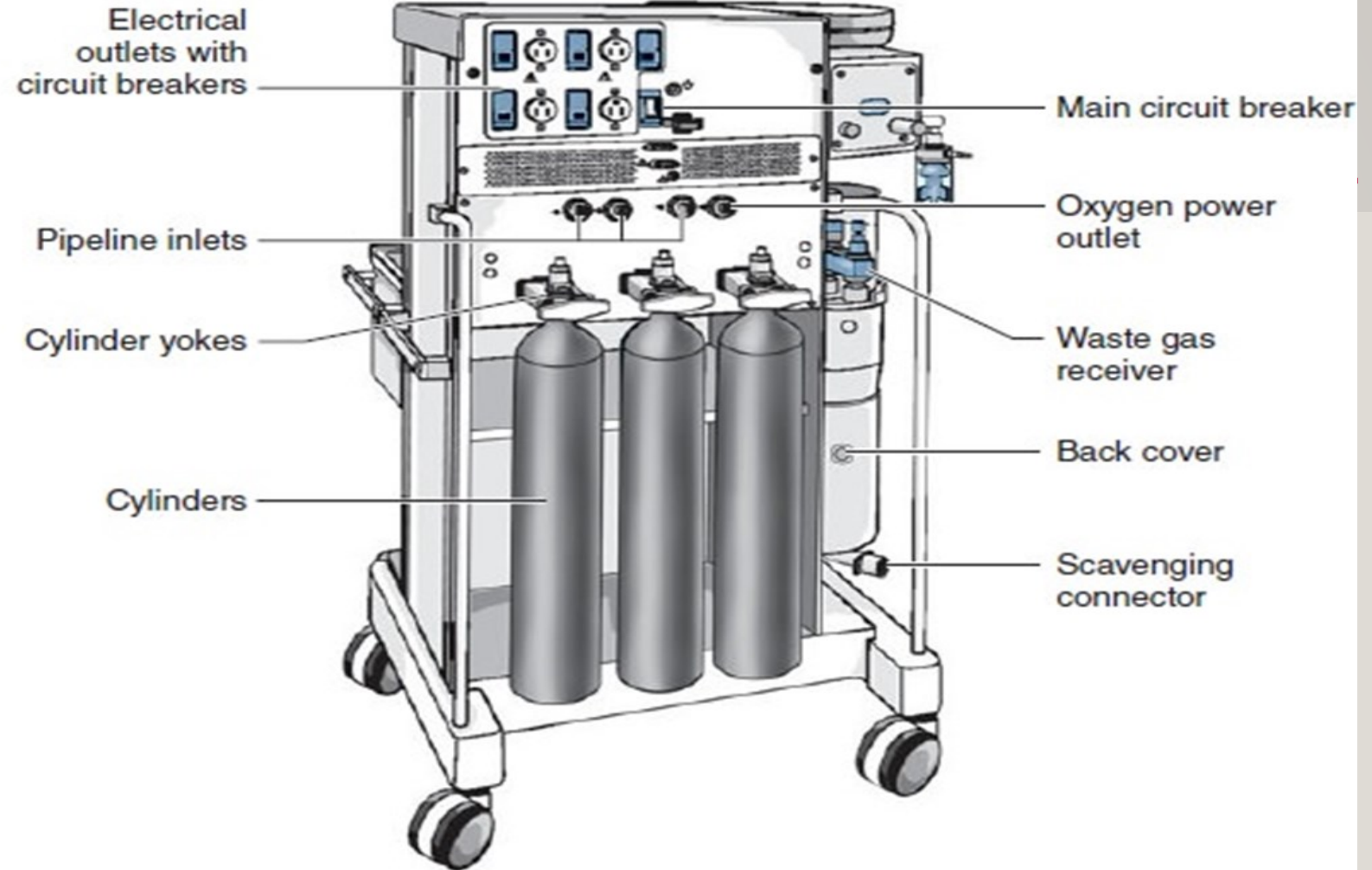


FIGURE 4-1 Modern anesthesia machine (Datex-Ohmeda Aestiva). A: Front. B: Back.

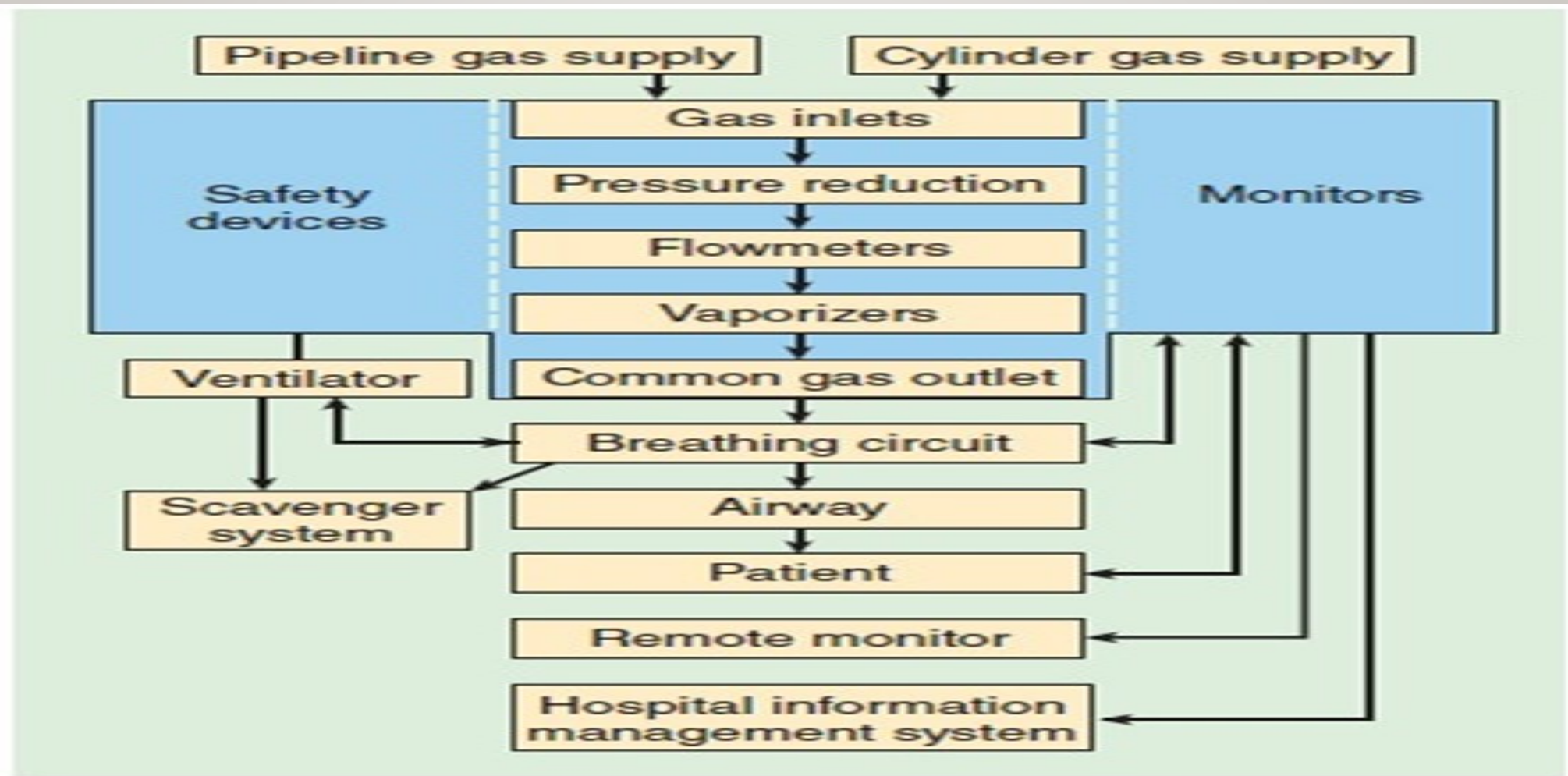


FIGURE 4-2 Functional schematic of an anesthesia machine/workstation.

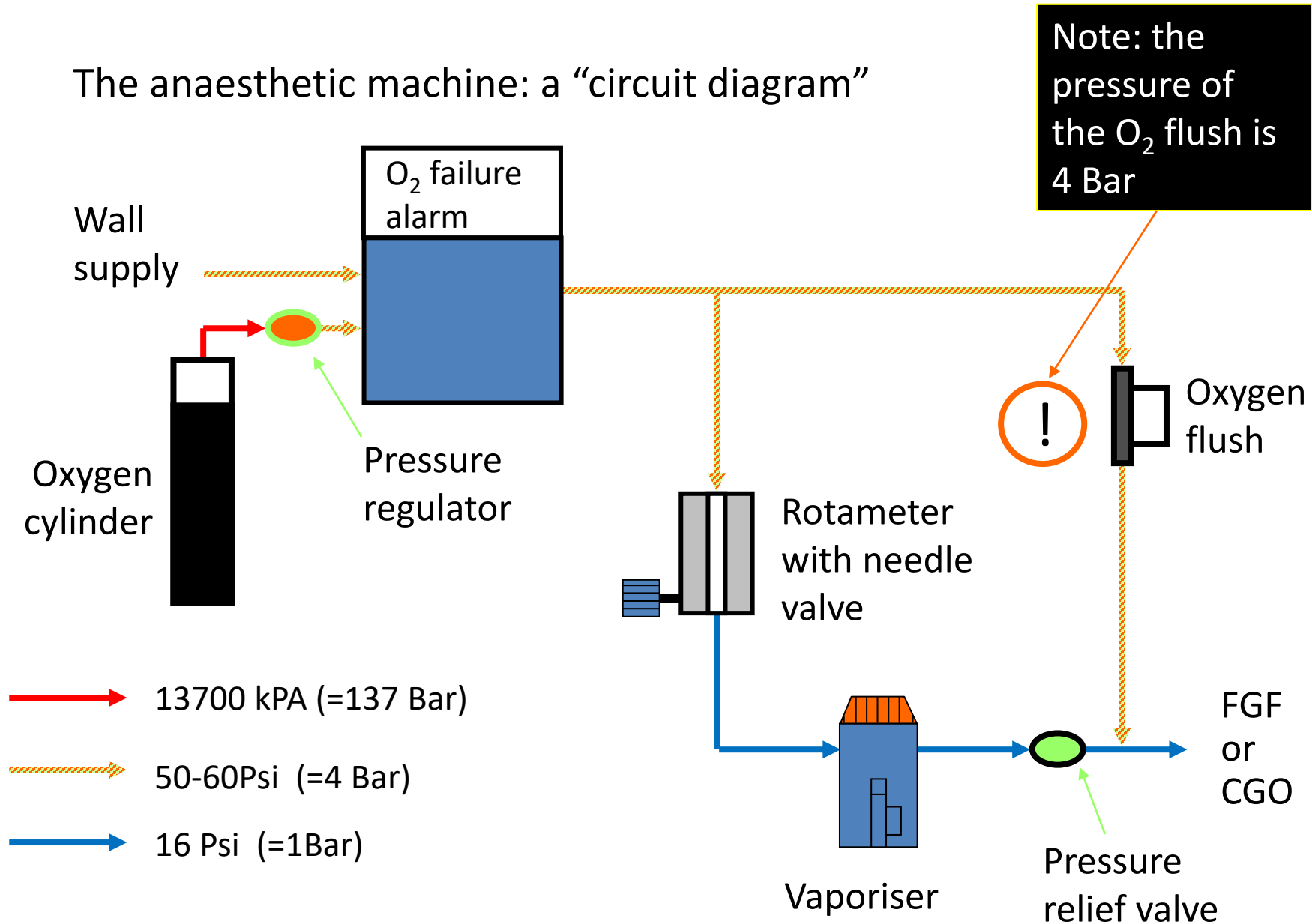
.....

- Cylinder pressure - 137 Bar; too high for the anesthetic machine.
- Reduced to 4 Bar to protect the machine
- After the rotameters, → slightly above atmospheric pressure (1.5 bar) to protect the PATIENT.
- This requires pressure regulators or pressure release valves.

GAS SUPPLY

- ✓ Most machines have gas inlets for oxygen, nitrous oxide, and air.
- ✓ Compact models often lack air inlets, whereas other machines may have a fourth inlet for helium, heliox, carbon dioxide, or nitric oxide.
- ✓ Separate inlets are provided for the primary pipeline gas supply that passes through the walls of health care facilities and the secondary cylinder gas supply.
- ✓ Machines therefore have two gas inlet pressure gauges for each gas: one for pipeline pressure and another for cylinder pressure.

The anaesthetic machine: a “circuit diagram”



PIPELINE INLETS

- Oxygen and nitrous oxide (and often air) are delivered from their central supply source to the operating room through a piping network.
- The tubing is color coded and connects to the anesthesia machine through a non interchangeable diameter-index safety system (DISS) fitting that prevents incorrect hose attachment.
- The non interchangeability is achieved by making the bore diameter of the body and that of the connection nipple specific for each supplied gas.
- A filter helps trap debris from the wall supply and a one-way check valve prevents retrograde flow of gases into the pipeline supplies.
- It should be noted that most modern machines have an oxygen (pneumatic) power outlet that may be used to drive the ventilator or provide an auxiliary oxygen flowmeter.
- The DISS fittings for the oxygen inlet and the oxygen power outlet are identical and should not be mistakenly interchanged.
- The approximate pipeline pressure of gases delivered to the anesthesia machine is 50 psig.

CYLINDER INLETS

- ✓ Cylinders attach to the machine via hanger-yoke assemblies that utilize a pin index safety system to prevent accidental connection of a wrong gas cylinder.
- ✓ The yoke assembly includes index pins, a washer, a gas filter, and a check valve that prevents retrograde gas flow.
- ✓ The gas cylinders are also color-coded for specific gases to allow for easy identification.
- ✓ In North America the following color-coding scheme is used: oxygen = green, nitrous oxide = blue, carbon dioxide = gray, air = yellow, helium = brown, nitrogen = black. In the United Kingdom, white is used for oxygen and black and white for air.

• • • • •

- A high-pressure source of medical gases and are generally used only as a back-up supply in case of pipeline failure.
- Pressure of gas supplied from the cylinder to the anesthesia machine is 45 psig.
- Some machines have two oxygen cylinders so that one cylinder can be used while the other is changed.
- At 20°C, a full cylinder contains 600 L of oxygen at a pressure of 1900 psig, and 1590 L of nitrous oxide at 745 psig.
- Cylinder pressure is usually measured by Bourdon pressure gauge .A flexible tube within this gauge straightens when exposed to gas pressure, causing a gear mechanism to move a needle pointer.

FLOW CONTROL KNOBS

- Flow control knobs control gas entry into the flowmeters by adjustment via a needle valve.
- Flowmeters on anesthesia machines are classified as either constant-pressure variable-orifice (rotameter) or electronic.
- In constant-pressure variable orifice flowmeters, an indicator ball, bobbin, or float is supported by the flow of gas through a tube.

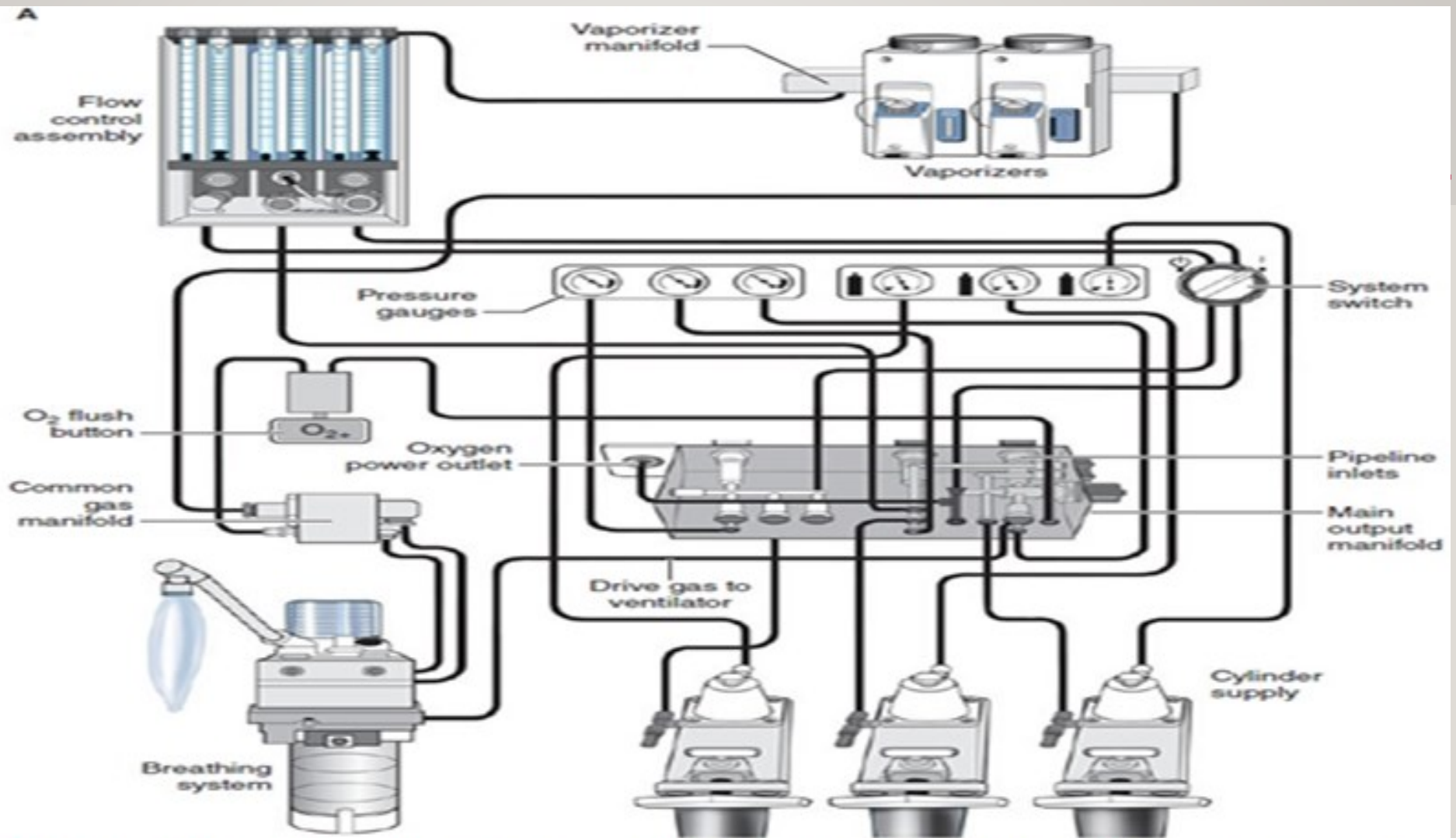


FIGURE 4-3 Simplified internal schematic of an anesthesia machine. **A:** Datex-Ohmeda Aestiva. (continued)

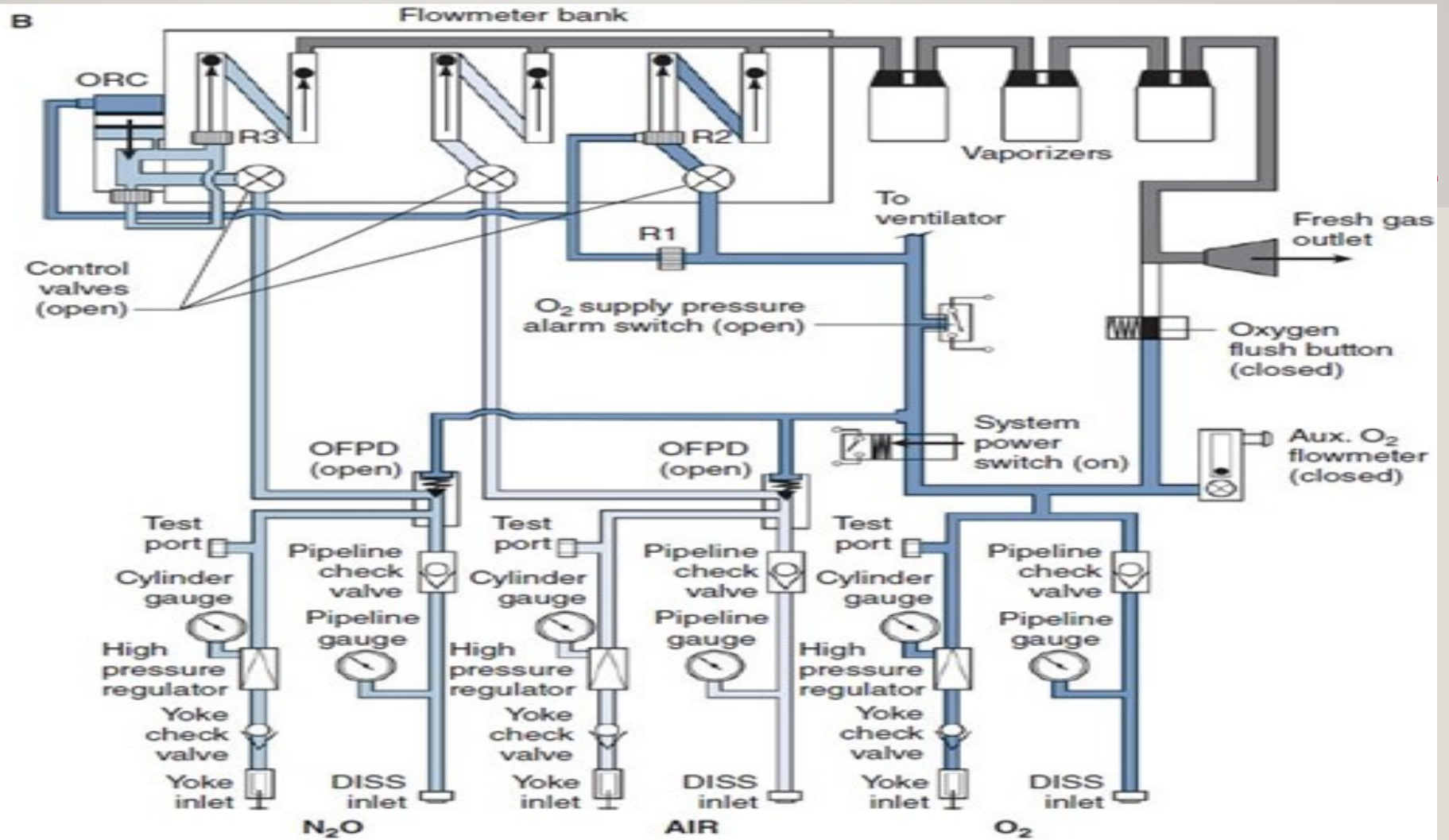
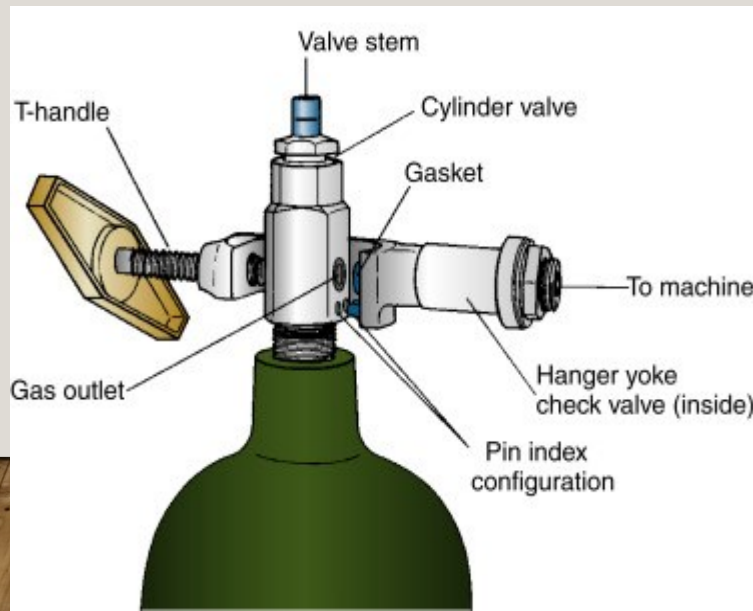


FIGURE 4-3 (continued) **B:** Dräger Narkomed. ORC, oxygen ratio controller.

HANGER YOKE ASSEMBLY

- The hanger yoke orients the cylinder, provides gas tight seal and ensures a unidirectional gas flow .
- The workstation standard requirement is that there should be at least one yoke for O₂ and N₂O.
- If the machine is likely to be used in locations that do not have piped gases, it is advisable to have a double yoke, especially for O₂.
- Machines are usually equipped with one or two E type cylinders that hang on specific hanger yokes



PIN INDEX SAFETY SYSTEM

The medical gas pin-index safety system ensures that the correct medical gas cylinder is hung in the correct yoke.

- The system consists of two pins that are fixed in the yoke, and which fit into two corresponding holes in the cylinder valve.
- The two pins are in a unique configuration for each gas and should never be removed from the hanger yoke. Specific pin configurations exist for each of the medical gases supplied in small cylinders in order to prevent erroneous misconnections of gas supplies.
- A cylinder should never be force-fitted to a hanger yoke
- For O₂ the pin index number is 2-5 and for N₂O it is 3-5.
- Substitution of an E type N₂O cylinder for O₂ can occur if pins in the index face are missing or broken or if several washers are used simultaneously.
- This fault is deceptive because the PISS is thought fool proof.
- This defect can only be detected by direct inspection of the yoke and cylinder gas identity markings each time a cylinder is replaced on the machine.

E Size Compressed Gas Cylinders

Cylinder Characteristics	Oxygen	Nitrous Oxide	Carbon Dioxide	Air
Color	White (green)	Blue	Gray	Black/White (yellow)
State	Gas	Liquid and gas	Liquid and gas	Gas
Contents (L)	625	1590	1590	625
Empty Weight (kg)	5.90	5.90	5.90	5.90
Full Weight (kg)	6.76	8.80	8.90	
Pressure Full (psig)	2000	750	838	1800

PRESSURE REDUCING DEVICE

- Reduces the high and variable pressure found in a cylinder to a lower and more constant pressure found in the anesthesia machine 45 psig
- Reducing device are preset so that the machine uses only gas from the pipeline(wall, gas),when the pipeline inlet pressure is 50 psig
- This prevents gas use from the cylinder even if the cylinder is left open i.e. saves the cylinder for backup if the wall gas pipeline fails
- Cylinders should be kept closed routinely .otherwise ,if the wall gas fails ,the machine will automatically switch to the cylinder supply without the anesthetist being aware that the wall supply has failed(until the cylinder is empty too)
- Modern anesthetic machine operates at pressure of 3-4 bar
- Hospital pipe line supplies also operate at a pressure of a 4 bar and therefore pressure regulators are not required between hospital pipeline and anesthetic machine

OXYGEN PRESSURE FAILURE DEVICE

- ✓ Machine standard requires that an anesthesia machine be designed so that whenever the oxygen supply pressure is reduced below normal, the oxygen concentration at the common gas outlet does not fall below 19%.
- ✓ A fail-safe valve is present in the gas line supplying each of the flow meter except oxygen
- ✓ This valve is controlled by the oxygen supply pressure and shuts off proportionally decrease the supply pressure of all other gases as the oxygen supply pressure decreases
- ✓ Historically there are two kind of fail safe valves
 1. Pressure sensor shut off valve (Ohmeda)
 2. Oxygen failure protection device (Drager)

• • • • •

- The machine standard specifies that whenever the oxygen supply pressure falls below a manufacturer- specified threshold (usually 30 psig)a medium priority alarm shall blow with in 5 seconds

Limitation of oxygen fail safe device alarm

- Failsafe valves do not prevent administration of a hypoxic gas mixture because they depend on pressure not flow
- These devices do not prevent hypoxia from accidents such as pipeline cross over or a cylinder containing the wrong gas

OXYGEN FLUSH VALVE

- Receives oxygen pipeline inlet or cylinder reducing device and directs high un metered flow directly to the common gas outlet downstream of the vapor
- The oxygen flush valve provides a high flow (35–75 L/min) of oxygen directly to the common gas outlet, bypassing the flowmeters and vaporizers.
- It is used to rapidly refill or flush the breathing circuit, but because the oxygen may be supplied at a line pressure of 45–55 psig, there is a real potential of lung barotrauma.
- For this reason, the flush valve must be used cautiously whenever a patient is connected to the breathing circuit.
- Moreover, inappropriate use of the flush valve (or a situation of stuck valve) may result in backflow of gases into the low-pressure circuit, causing dilution of inhaled anesthetic concentration.
- Some machines use a second-stage regulator to drop the oxygen flush pressure to a lower level.

Hazards

- It may cause barotrauma
- It may cause dilution of inhaled anesthetic



Adjustable Pressure-Limiting Valve

- The adjustable pressure-limiting (APL) valve, some-times referred to as the pressure relief or pop-off valve, is usually fully open during spontaneous ventilation but must be partially closed during manual or assisted bag ventilation.
- The APL valve often requires fine adjustments. If it is not closed sufficiently excessive loss of circuit volume due to leaks prevents manual ventilation.
- At the same time if it is closed too much or is fully closed a progressive rise in pressure could result in pulmonary barotrauma (e.g., pneumothorax) or hemodynamic compromise, or both.
- As an added safety feature, the APL valves on modern machines act as true pressure-limiting devices that can never be completely closed; the upper limit is usually 70–80 cm H₂O.

HUMIDITY

- Absolute humidity is defined as the weight of water vapor in 1 L of gas (ie, mg/L).
- Relative humidity is the ratio of the actual mass of water present in a volume of gas to the maximum amount of water possible at a particular temperature.
- At 37°C and 100% relative humidity, absolute humidity is 44 mg/L, whereas at room temperature (21°C and 100% humidity) it is 18 mg/L.
- Inhaled gases in the operating room are normally administered at room temperature with little or no humidification.
- Gases must therefore be warmed to body temperature and saturated with water by the upper respiratory tract. Tracheal intubation and high fresh gas flows bypass this normal humidification system and expose the lower airways to dry (<10 mg H₂O/L), room temperature gases.

• • • • •

- Prolonged humidification of gases by the lower respiratory tract leads to dehydration of mucosa, altered ciliary function, and, if excessively prolonged, could potentially lead to inspissation of secretions, atelectasis, and even ventilation/perfusion mismatching, particularly in patients with underlying lung disease.
- Body heat is also lost as gases are warmed and even more importantly as water is vaporized to humidify the dry gases.
- The heat of vaporization for water is 560 cal/g of water vaporized. Fortunately, this heat loss accounts for about only 5–10% of total intraoperative heat loss, is not significant for a short procedure (<1 h), and usually can easily be compensated for with a forced-air warming blanket.
- Humidification and heating of inspiratory gases may be most important for small pediatric patients and older patients with severe underlying lung pathology, e.g., cystic fibrosis.

BASIC COMPONENT OF THE ANESTHESIA MACHINE

- Source of compressed gases ,Pipeline Supply and Cylinder Supply
- Vaporizers; Volatile anesthetics must be vaporized before being delivered to the patient.
- Flow meters ;The flow meter assembly precisely controls and measures gas flow to the common gas outlet.
- Bellows ;are analogues to the reservoir bag in the breathing circuit and they act an interface between breathing system gas and ventilator driving gas.
- Ventilator; The ventilator on the modern anesthesia workstation serves as a mechanized substitute for the hand of the anesthesia care provider
- Scavenging; is the collection and subsequent removal of waste anesthetic gases from the operating room



OXYGEN CYLINDER



FLOW METER ASSEMBLY

- When the flow control valve is opened the gas enters at the bottom and flows up the tube elevating the indicator
- The indicator floats freely at a point where the downward force on it (gravity) equals the upward force caused by gas molecules hitting the bottom of the float



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VENTILATOR

- Ventilators generate gas flow by creating a pressure gradient between the proximal airway and the alveoli.
- Modern ventilators generate positive pressure and gas flow in the upper airway.
- Ventilator function is best described in relation to the four phases of the ventilatory cycle: inspiration, the transition from inspiration to expiration, expiration, and the transition from expiration to inspiration.
- Although several classification schemes exist, the most common is based on inspiratory phase characteristics and the method of cycling from inspiration to expiration.

VAPORIZERS

- ❖ An anesthetic vaporizer or anesthetic vaporizer is a device generally attached to an anesthetic machine which delivers a given concentration of a volatile anesthetic agent.
- ❖ Volatile anesthetics (e.g., halothane, isoflurane, desflurane, sevoflurane) must be vaporized before being delivered to the patient.
- ❖ Vaporizers have concentration-calibrated dials that precisely add volatile anesthetic agents to the combined gas flow from all flowmeters.
- ❖ They must be located between the flow-meters and the common gas outlet. Moreover, unless the machine accepts only one vaporizer at a time, all anesthesia machines should have an interlocking or exclusion device that prevents the concurrent use of more than one vaporizer.

PRECISION VAPORIZER

- Flow compensated
- Temperature compensated
- High resistance
- Known concentration
- Agent specific

VAPORIZERS



CANISTER

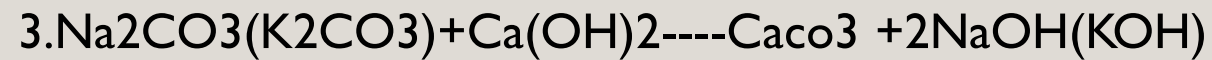
- The typical anesthesia machine absorber is **a clear plastic canister or collection of canisters of varying sizes but at least of the same volume as a patient's typical tidal volume**. The canister is full of some type of alkaline absorbent granules.

Soda lime mostly commonly used

- 80% calcium hydroxide ,15% water,4% sodium hydroxide and 1%potassium hydroxide (an activator)
- Silica

ABSORPTION

The equations



Baraylime 20% barium hydroxide and 80% calcium hydroxide

Calcium hydroxide lime

lack of sodium and potassium hydroxide

Carbon monoxide and the nephrotoxic substance known as compound A

CANISTER



ANESTHETIC CIRCUIT

- A breathing system or breathing circuit is a medical device used to deliver oxygen and anesthetic gases to the patient.
- It eliminate carbon dioxide, gives adequate in flow of fresh gas and Carbon dioxide absorbent in Semi closed rebreathing circuits and circle system.

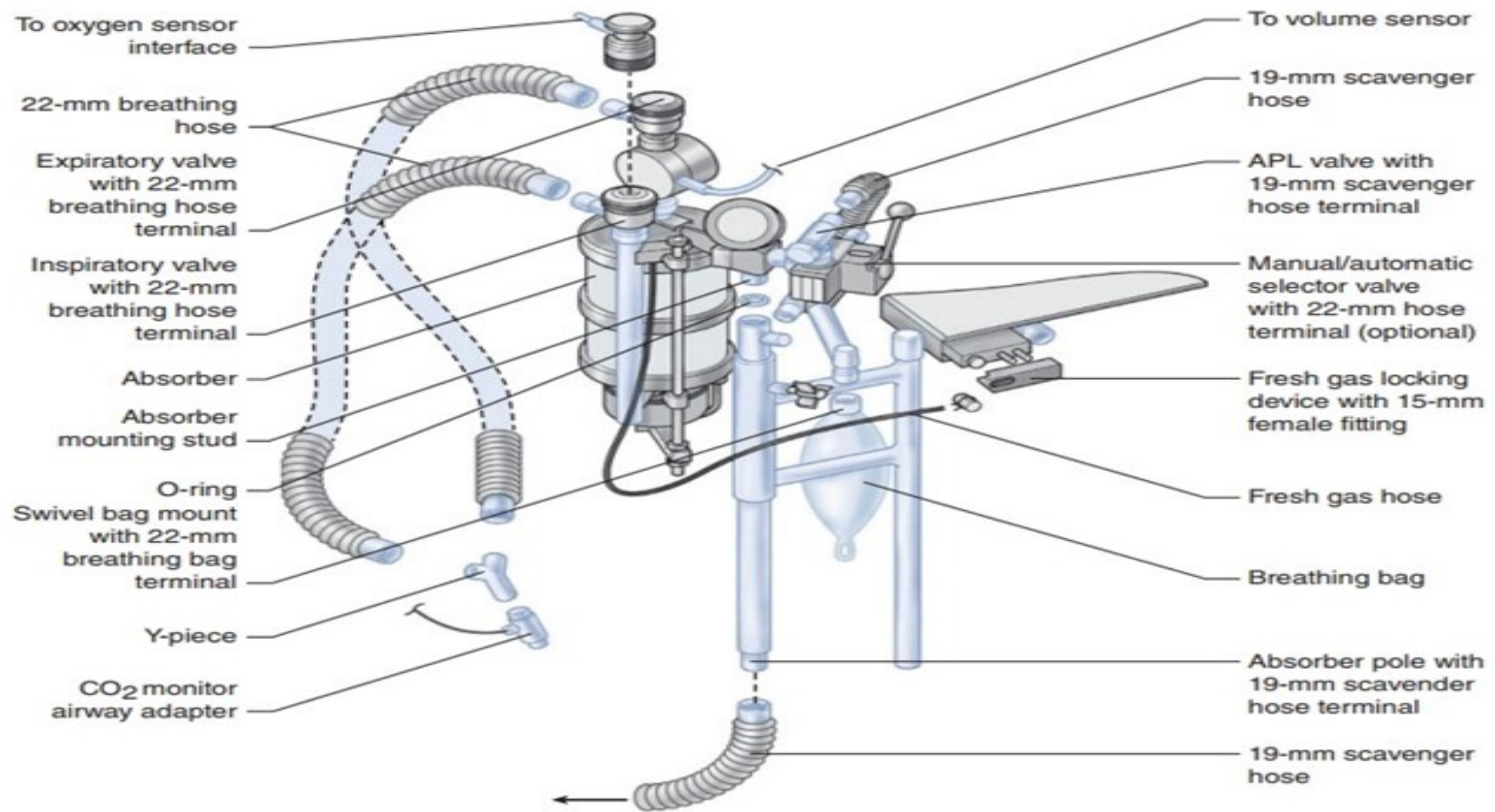


FIGURE 4-20 Breathing circuit design. **A:** Conventional external components. (continued)

CHECKING ANESTHESIA MACHINE

categories of check

- Emergency ventilation equipment
- High pressure system
- Low pressure system
- Breathing system
- Manual and automatic system
- Monitors and final position

Checking Anaesthetic Equipment 2012

Check self-inflating bag available

Perform manufacturer's (automatic) machine check

Power supply

- Plugged in
- Switched on
- Back-up battery charged

Gas supplies and suction

- Gas and vacuum pipelines – 'tug test'
- Cylinders filled and turned off
- Flowmeters working (if applicable)
- Hypoxic guard working
- Oxygen flush working
- Suction clean and working

Breathing system

- Whole system patent and leak free using 'two-bag' test
- Vaporisers – fitted correctly, filled, leak free, plugged in (if necessary)
- Soda lime - colour checked
- Alternative systems (Bain, T-piece) – checked
- Correct gas outlet selected

Ventilator

- Working and configured correctly

Scavenging

- Working and configured correctly

Monitors

- Working and configured correctly
- Alarms limits and volumes set

Airway equipment

- Full range required, working, with spares

RECORD THIS CHECK IN THE PATIENT RECORD

Don't Forget!

- Self-inflating bag
- Common gas outlet
- Difficult airway equipment
- Resuscitation equipment
- TIVA and/or other infusion equipment

CHECKS BEFORE EACH CASE

Breathing system

Whole system patent and leak free using 'two-bag' test
Vaporisers – fitted correctly, filled, leak free, plugged in (if necessary)
Alternative systems (Bain, T-piece) – checked
Correct gas outlet selected

Ventilator

Working and configured correctly

Airway equipment

Full range required, working, with spares

Suction

Clean and working

THE TWO-BAG TEST

A two-bag test should be performed after the breathing system, vaporisers and ventilator have been checked individually

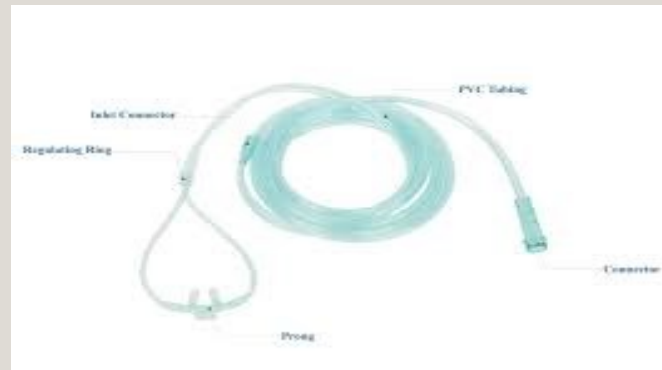
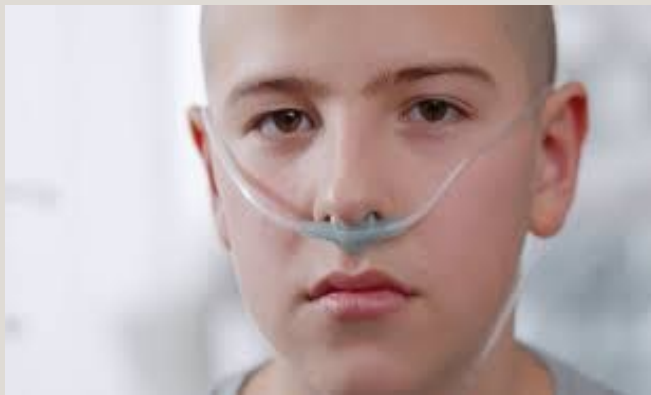
- i. Attach the patient end of the breathing system (including angle piece and filter) to a test lung or bag.
- ii. Set the fresh gas flow to 5 L.min^{-1} and ventilate manually. Check the whole breathing system is patent and the unidirectional valves are moving. Check the function of the APL valve by squeezing both bags.
- iii. Turn on the ventilator to ventilate the test lung. Turn off the fresh gas flow, or reduce to a minimum. Open and close each vaporiser in turn. There should be no loss of volume in the system.

AIR WAY EQUIPMENT



AIR WAY EQUIPMENT

- Oxygen delivery system
- Nasal canula ;
- Flow 1-6 l/min
- Fio2 25-40%



FACE MASK

- A simple face mask can deliver 35% to 50% oxygen with an appropriate flow rate of 6 to 10 L/minute.
- A minimum of 6 L/minute of oxygen flow is needed to prevent rebreathing of exhaled carbon dioxide.
- A partial rebreathing mask with a reservoir bag is a face mask that delivers moderate to high concentrations of oxygen.

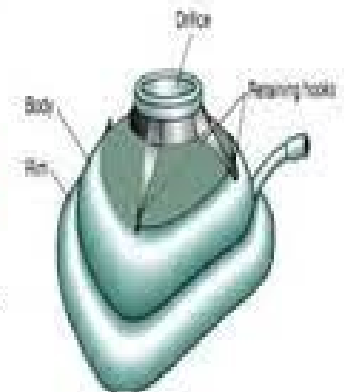
FACE MASK



PARTS OF FACEMASK

Body (Shell or Dome)

- Constitutes the main part of the mask
- **Transparent facemask** allows observation for vomitus, secretions, blood, lip color, and exhaled moisture and is better accepted by a conscious patient.



FACE TENT

- A face tent is a shieldlike device that fits under the patient's chin and encircles the face. It is used primarily for humidification and for oxygen only when the patient cannot or will not tolerate a tight-fitting mask



NON REBREATHER

- Non-rebreather mask is a device used in medicine to assist in the delivery of oxygen therapy. A NRB requires that the patient can breathe unassisted, but unlike a low-flow nasal cannula, the NRB allows for the delivery of higher concentrations of oxygen.



Oropharyngeal Airway

- Indications

- OP Airways are indicated in unconscious patients without a gag reflex
- An Oral Airway will prevent the tongue, which is the most common airway obstruction, from blocking the airway when properly inserted.

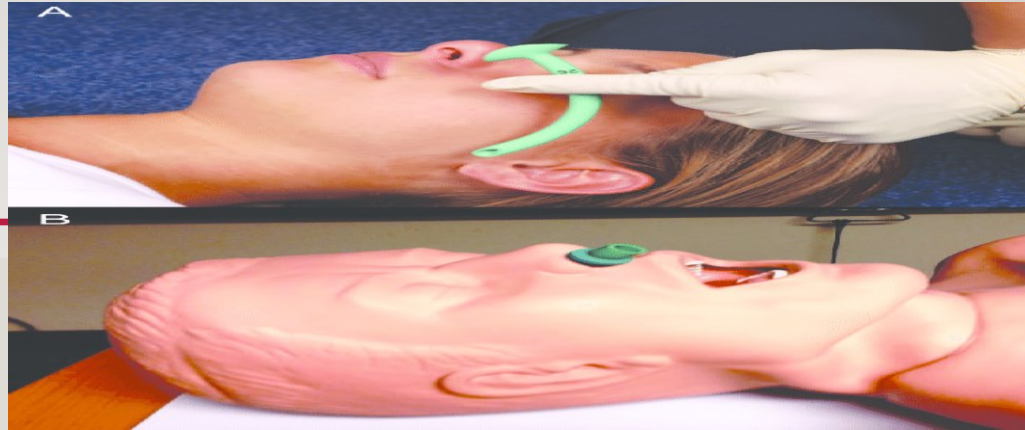


Oropharyngeal airway (guedel)

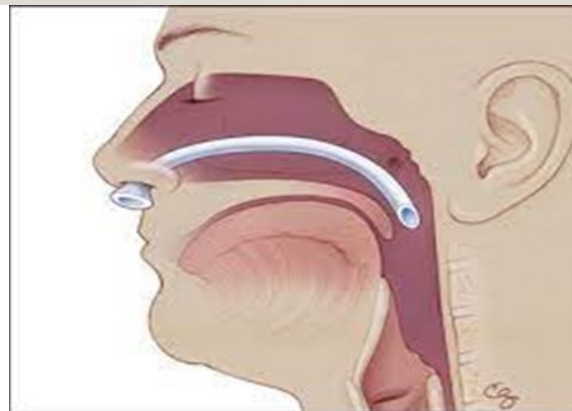
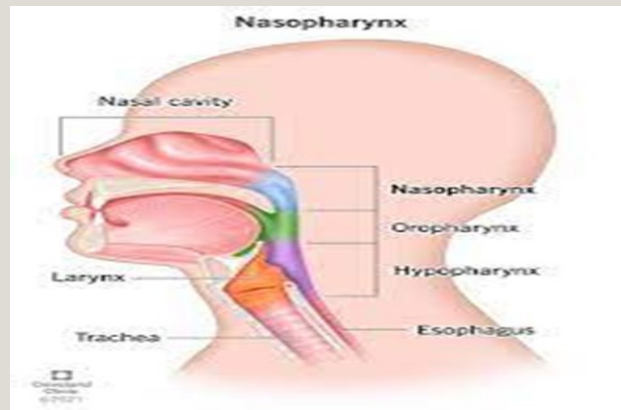
- Curved plastic tubes, flattened in cross-section and flanged at the oral end. They lie over the tongue, preventing it from falling back into the pharynx.



NASAL AIRWAY



A **nasopharyngeal airway (NPA)** is a thin, clear, flexible tube that is inserted into a patient's nostril. The purpose of the NPA is to bypass upper airway obstruction at the level of the nose, nasopharynx or base of the tongue.



LARYNGOSCOPE

- Laryngoscopes are used mainly by anesthetists to expose a patient's trachea opening for ease of tracheal intubation to maintain efficient ventilation.
- It is one of the essential tools for enabling anesthetists to secure the airway and care for anesthetized or critically ill patients.
- Laryngoscopes are designed for visualization of the vocal cords and for placement of the ETT into the trachea under direct vision.
- The two main types are the curved Macintosh blade and the straight blade (i.e., Miller with a curved tip and Wisconsin or Foregger with a straight tip).

View of the larynx at laryngoscopy

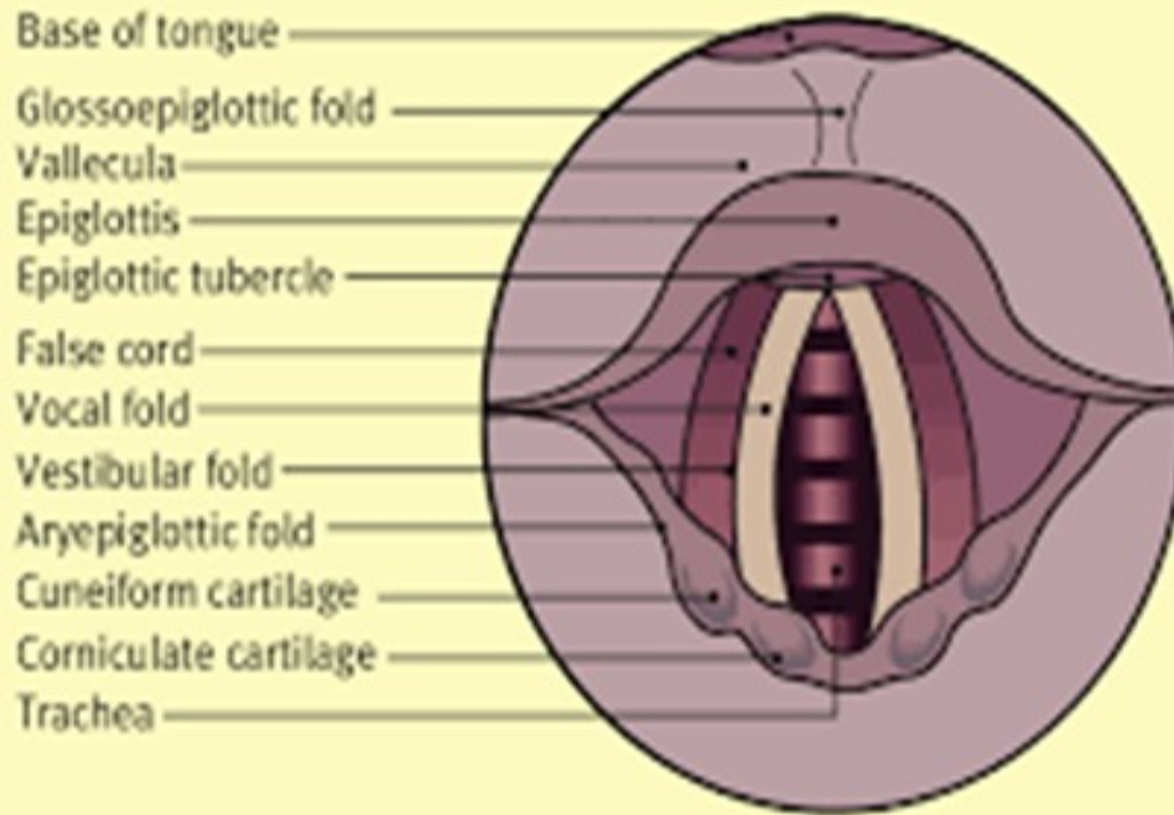
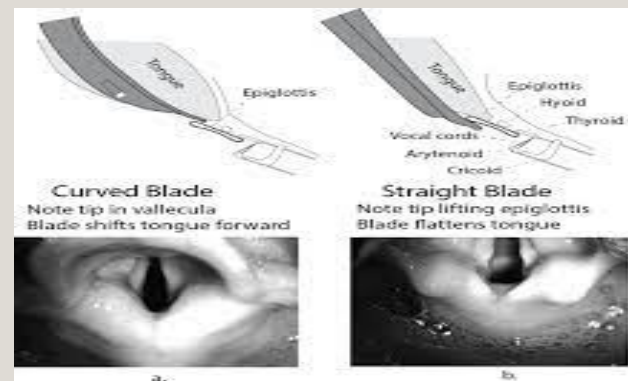
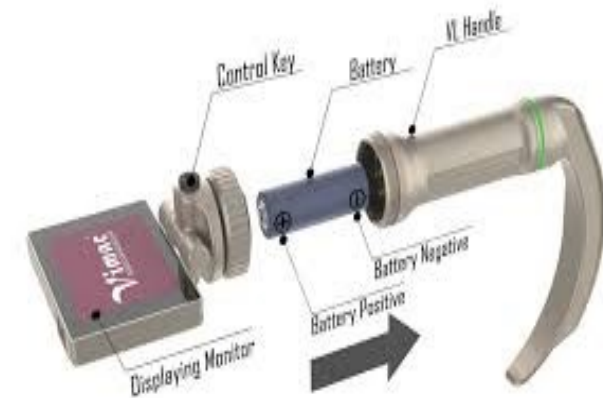
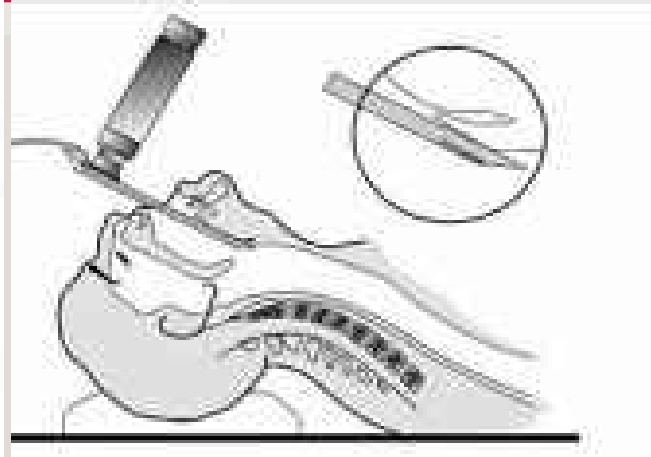


Figure 3



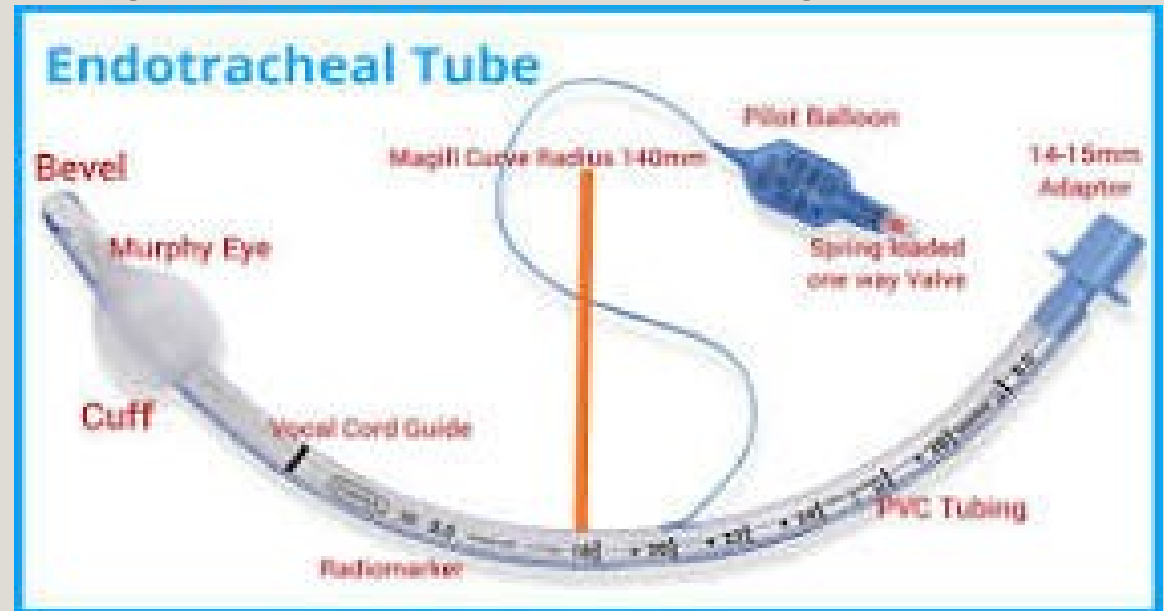
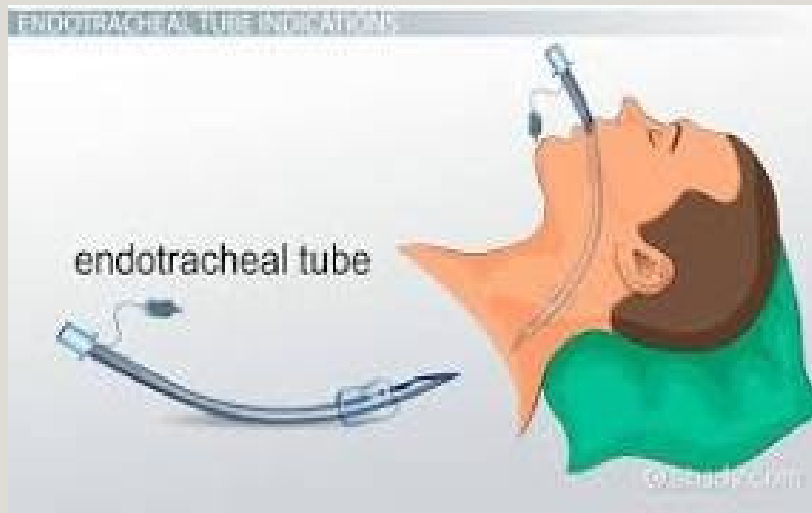
LED Laryngoscope Set

LARYNGOSCOPE



ENDOTRACHEAL TUBE

In its simplest form, the endotracheal tube is a tube constructed of polyvinyl chloride that is placed between the vocal cords through the trachea. It serves to provide oxygen and inhaled gases to the lungs and protects the lungs from contamination, such as gastric contents or blood.



LARYNGEAL MASK AIRWAY

- The laryngeal mask airway (LMA) is a supraglottic airway device developed by British Anesthesiologist Dr. Archi Brain. It has been in use since 1988. Initially designed for use in the operating room as a method of elective ventilation, it is a good alternative to bag-valve-mask ventilation, freeing the hands of the provider with the benefit of less gastric distention.
- Initially used primarily in the operating room setting, the LMA has more recently come into use in the emergency setting as an important accessory device for management of the difficult airway.
- The LMA is shaped like a large endotracheal tube on the proximal end that connects to an elliptical mask on the distal end. It is designed to sit in the patient's hypopharynx and cover the supraglottic structures, thereby allowing relative isolation of the trachea



BOUGIE AND STYLET

- A stylet is a malleable metal rod placed inside the endotracheal tube to facilitate its passage into the trachea.
- A bougie is a thin plastic rod that is passed into the trachea, over which the endotracheal tube is inserted

(a)



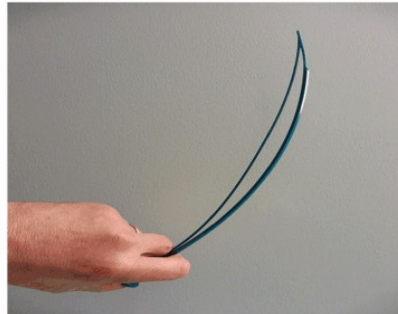
(b)



(c)



(d)



(e)



AIR WAY EQUIPMENT'S

- Medical Gas supply
- Anesthetic machine
- Humidification and filtration
- Breathing systems
- Airways
- Tracheal intubation equipment
- Masks and oxygen delivery devices
- Ventilators



The end

